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Portable Semantic Models and Ad Hoc Reconciliation by Cognitive Agents in Network Management

Abstract

Interoperation between network-management systems has traditionally required a data model agreed in advance. Where that is insufficient, a richer shared information model or ontology is agreed instead. Both demand broad prior agreement, negotiated and maintained by people. This document examines what becomes possible once the systems on each side can reason. Each side can lift its own data into a complete semantic model. Two such models can then reconcile ad hoc, for the occasion, with no model agreed beforehand.

The document develops the semantic model and the two operations upon it: the lift that produces one model, and the reconciliation that bridges two. It gives careful treatment to pragmatics, the layer a schema is least able to hold and so the part most easily left out of view, yet often the one that carries key operative meaning. It considers the form and utility of thin references, separately assessing the two distinct roles they may play. It then reports an empirical study across four network-management scenarios, run over independently constructed cases and several cognitive-agent families. The study measures the portability of a lifted model, the extent to which cognition can reconcile divergent models without support of a pre-agreed standard, where a reference is load-bearing and what it usefully comprises. It closes with conclusions for how information should be prepared for cognitive systems, and for how the community's standardization effort might evolve.

This is a research document intended to inform discussion in the IRTF Network Management Research Group. It reflects the authors' ideas, thoughts and experimental findings and does not represent IETF or IRTF consensus.

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1. Introduction

Network management is acquiring participants that reason. Autonomous systems will consume management information, act on it, and exchange it with one another. They will bring machine cognition to a plane whose reasoning and design have, until now, lain with human engineers. The information those systems are handed was shaped for particular programs and the junctions between them, not for stand-alone comprehension.

Two software systems that must exchange information rarely share a model of the world. The classical remedy is a data model agreed in advance. Where that falls short, a richer shared information model or ontology is agreed. Both require broad prior agreement on a common model, negotiated and maintained by people. Both are slow and costly to reach. This document takes up the alternative that opens once the systems on each side can reason for themselves. They can reconcile their divergent models *ad hoc*, for the occasion, machine to machine, with nothing settled beforehand.

The central question is therefore not how to write a better shared data model. It is whether any advance agreement is still required or useful once the participants can reason, and, where it may be, in what form. This document argues, and supports with measurement, that a great deal of what was formerly pre-agreed can instead be produced and reconciled on demand by cognition. What remains worth agreeing is thinner, and more broadly useful, than a full shared model.

The account rests on one object and two operations, kept carefully distinct throughout:

- the **semantic model**: the complete, meaning-bearing representation a system's data is lifted into, so that a machine can comprehend it without a human supplying what a schematic form leaves out;
- the **lift**: the cognitive act that produces one semantic model from a system's data; and
- **reconciliation**: the cognitive operation that bridges two independently produced semantic models, so that the systems behind them can interoperate.

The semantic model is the output of the lift and the input to reconciliation. The two operations are easily bundled together, yet they are distinct. Completing a single model is needed even for one system in isolation, simply as the price of feeding a cognitive consumer. Reconciling two is needed only when two systems must meet. This document treats them separately, then relates them.

The contributions are four. First, a self-contained account of the semantic model, the lift, and reconciliation for a network-management audience ([Section 3](#), [Section 4](#), [Section 6](#)). Second, a focused treatment of **pragmatics**, the layer that most often carries the operative meaning and is least often discussed ([Section 5](#)). Third, a clear separation of the two distinct roles of a thin shared **reference** ([Section 7](#)). Fourth, an empirical study across four network-management scenarios that measures portability, reconciliation without a pre-agreed standard, and where a reference is load-bearing ([Section 8](#)). The document closes with the conclusions the evidence supports ([Section 9](#)) and their implications for standardization ([Section 10](#)).

This document stands on its own. Three earlier Internet-Drafts by one of the authors explored these ideas at an earlier and less mature stage ([Section 11](#)). They are cited as related work and are not assumed here.

2. Terminology and Conventions

This is an Informational research document. It defines no protocol and uses no normative keywords. The following terms are used with the specific meanings given here.

Data model

A syntactic and structural artifact, for example a YANG module or a JSON schema, that fixes a form. It carries meaning only implicitly, through the structures it prescribes.

Ontology

An explicit account of what a domain admits as entities, attributes, and relationships. It concerns what exists and how things relate, not how any schema expresses it.

Lexicon

The set of terms by which an ontology's entities, attributes, and relationships are named. A lexicon is a feature of an ontology, not a separate layer beneath it.

Knowledge graph

The ontology together with the individuals that populate it: the typed concepts and permitted relations (the schema, or TBox) and the asserted instances (the data, or ABox). A lifted semantic model contains a knowledge graph, and adds pragmatics and provenance.

Pragmatics

How the content is to be used: what a thing is *for*, whose judgment governs it, whether a figure is a request or a measurement, whether a state is intended or actual, and what a signal means in a given operating context. Pragmatics is the part a schema is least able to hold. It is treated at length in [Section 5](#).

Provenance

The base recording who asserted a given fact, by what method, and how firmly. It runs across all of the above.

Semantic model

What a representation needs in order to be *complete*, in the specific sense that a cognitive system can comprehend it and act on it without a human supplying meaning the form leaves out. Concretely: an ontology, including its lexicon, together with pragmatics, carried over a base of provenance.

Lift

The cognitive act of producing a semantic model from a system's data, making explicit the meaning a data model carries only implicitly.

Portability

The property of a soundly produced semantic model whereby any sufficiently capable cognitive agent can pick it up and use it, from the package alone, with no pre-agreed standard. Portability is a property of the model, not of any particular downstream task.

Reconciliation

The operation of establishing, meaning element by meaning element, how two independently produced semantic models correspond, closing the semantic gaps between them, so that the systems that use them natively can interoperate.

Reference

A thin, shared, consultable nomenclature (names, disambiguating definitions, and examples) to which a lift or a reconciliation can tie terms and meanings. It is not an ontology or a data schema. Its two distinct roles are the subject of [Section 7](#).

False cognate

Two terms that share a label but denote unrelated things. A disciplined reconciliation must refuse such a pairing rather than bind it.

Cognitive agent

An autonomous software entity built around a language model together with controlled stores of knowledge, able to reason about what it is given and to communicate in natural language.

Comprehension score

The measure of portability used in [Section 4](#): the share of a lifted model's concepts that an independent agent, given only the model package, explains faithfully. Each explanation is graded by meaning, against an authored answer key, as faithful, partial, an honest abstention (the meaning cannot be determined from the package), or invented (a meaning the key does not support). The comprehension score is the fraction judged faithful.

Resolved fraction; precision

Two measures of a reconciliation, used throughout [Section 8](#). The *resolved fraction* is the share of the true correspondences that an agent finds and commits. *Precision* is the share of what it commits that is correct.

3. The Central Object: the Semantic Model

Everything turns on the model a system's data is lifted into. A data model gives form and some of the denotation. What it defers to specification, to convention, and to the engineer's own knowledge are gaps to fill. The semantic model supplies that remainder. It is **semantic**: it carries meaning, not merely structure. It is **ad hoc**: built for the case at hand, with no standard model agreed in advance. And it is **complete**: it carries, as far as possible, the two layers a bare schema generally does not.

Those two layers are pragmatics and provenance, and they lie over and above the ontology and its lexicon. The ontology and lexicon fix what the content is about and how its terms are named. Pragmatics governs how the content is to be used. It is the part a schema is least able to hold, and, as [Section 5](#) shows, often the one that carries important operative meaning. Provenance records who asserted each fact, by what method, and how firmly.

3.1. Grounding: the theory of signs

This structure is not invented for the occasion. It follows the long-established study of signs [[Morris](#)], which distinguishes three aspects of any representation. *Syntax* is its form, the rules by which it is well constructed. *Semantics* is what its parts denote. *Pragmatics* is its interpretative context: how it is used. The correspondence to the vocabulary here is direct. Syntax is form, which the semantic model sets aside, since the same meaning may be written in many forms. Semantics, what the parts denote, is carried by the ontology and its lexicon. And pragmatics keeps its name. Data models are strong on syntax and carry some semantics. What they say very little about, in a form a machine can act on, is pragmatics. Pragmatics is a large part of what a human engineer silently supplies when reading a model.

As an aside, the same three parts arrive from a different starting point. Placed on the data-to-wisdom ladder [[Ackoff](#)], data are values carried by a data model. Information is data placed in relation and given reference, which fixing an ontology and its lexicon provides. Knowledge is information organized to support inference and action, reached by adding pragmatics. The semantic model, ontology and lexicon plus pragmatics, is therefore *knowledge* in these terms. Wisdom, the top of the ladder, is not a further document at all. It is cognition itself, the faculty

that consumes and generates knowledge. On this reading a data model delivers data, the lift carries data up through information to knowledge, and the cognitive system supplies the wisdom that acts on it. The two framings align.

3.2. One object, two operations

Two operations must be separated at the outset, because they are easily bundled and are in fact distinct operations around the same objects. Completing a representation, producing a semantic model at all so a machine can comprehend it, is one thing. It is needed even for a single cognitive system in isolation. Reconciling two such models, making them mutually comprehensible, is another. It is needed only when two systems must meet. Figure 1 shows the relationship. The lift produces one semantic model from a system's data. Reconciliation consumes two, and, where the systems must then exchange native data, yields data-model translators as artifacts of the single reconciled result.

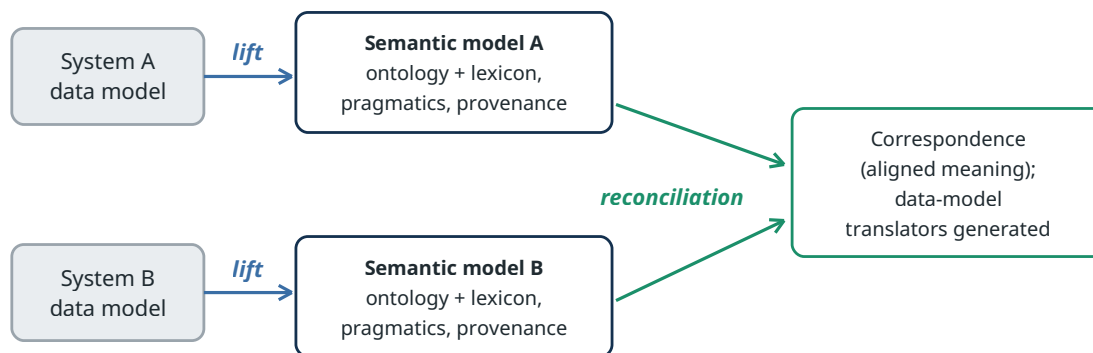


Figure 1: The semantic model and its two operations: the lift produces one; reconciliation bridges two.

The remainder of the document follows this division. First producing the model (Section 4 and Section 5), then reconciling two of them (Section 6), then the reference that assists both (Section 7).

4. Producing the Model: the Lift, and Portability

The lift turns a system's data into a complete semantic model. It is the first thing asked of machine cognition, and it is a cognitive act in its own right, not a schema transformation. It has a bar to clear. A lift is **adequate** when a sufficiently capable cognitive agent can pick the resulting model up and apply it to any end, from the package alone. A model that clears that bar is **portable**.

A semantic model is a set of lifted concepts, each carried under a short header. It helps to see a single such concept in full before measuring what a reader can do with the whole model. The concept below is drawn from the configuration scenario, an ONF Transport API connectivity-service, and it shows the parts a lift makes explicit:

```
{ "id": "t.cs",
  "label": "connectivity-service",
  "synonyms": ["service"],
  "kind": "service",
  "gloss": "an end-to-end connectivity service across the network",
  "example": "the A1-A3 ODU2 service",
  "relations": [ { "rel": "uses", "target": "t.sip"},
                  { "rel": "realized-by", "target": "t.cep"},
                  { "rel": "over", "target": "t.topo"} ],
  "instances": ["cs-a1a3-odu2 (ODU2, A1 to A3)",
  "ref": "connection-service",

  "pragmatics": {
    "use": "the billable, SLA-bearing unit of connectivity",
    "authority": "owned by the service layer; the transport side
                  may not redefine its grade" },

  "provenance": {
    "asserted-by": "the ONF TAPI controller",
    "method": "context export",
    "firmness": "authoritative" } }
```

Figure 2: One lifted concept: the ontology and lexicon, instances, pragmatics and provenance.

Read top to bottom, the concept separates cleanly into the layers of [Section 3](#). The label and synonyms are the lexicon. The kind, relations, and instances are the ontology, and they are exactly the typed schema (the TBox) and the populated data (the ABox) that make up the knowledge graph a lifted model contains. The gloss, example, and ref are supports for a reader rather than model content: the canonical example illustrates the concept for a person and is not one of the instances that populate the ABox. The last two blocks are the ones bare schema normally do not hold. The **pragmatics** record what the concept is for and whose judgment governs it, so that a consumer knows the service is the billable unit and that the transport side may not redefine its grade. The **provenance** records who asserted the concept, by what method, and how firmly.

4.1. The Semantic Model in a Standard Ontology Language

OWL and RDF can carry the full semantic model. The structural layers map natively, as [Table 1](#) shows; pragmatics rides as annotation, and provenance has a dedicated vocabulary, PROV-O [[W3C.PROV-O](#)]. Our JSON holds no representational advantage; the two are alternative serializations of the same content.

Lifted-concept field	OWL / RDF
concept	owl:Class
label	rdfs:label
synonyms	skos:altLabel
kind	a class annotation
gloss / example	skos:definition / skos:example
ref	skos:closeMatch to a reference entry
relations	typed object properties
instances	owl:NamedIndividual, typed to the concept

Table 1: A lifted concept's structural fields, mapped to OWL and RDF.

What determines whether the pragmatic layer is usable is not the serialization but how the model is consumed. A formal, deductive reasoner over either form can act on classes and individuals, yet it can only treat a pragmatic rule such as "recomputed from the current context, lower during a maintenance window" as inert annotation, since deduction cannot resolve a defeasible, context-relative judgment. A cognitive reader of either form resolves that same rule, reading it against the live operating context. The two serializations are interchangeable here; the mode of consumption is not.

The deeper point is that pragmatics is the content's context of use. Handling it is therefore nothing other than resolving context of use, which only a reader that reasons about context can do. What makes the pragmatic layer tractable is the kind of reader, not the serialization: a model consumed by cognition rather than by deduction. The semantic model is thus expressible in the community's existing knowledge-graph standards [W3C.OWL2] [W3C.RDF11] [W3C.SKOS], and what the approach adds is the reader that resolves its pragmatics.

4.2. Portability

Portability is worth isolating because it can be assessed directly. The test hands a lifted model to an independent consumer agent, one that never saw the source system and holds no inside knowledge of it. The agent explains each concept in its own words, from the package alone, and may answer that a concept cannot be determined from what it was given, since an honest abstention is a valid response and not a forced guess. A separate, fixed judge then grades each explanation by meaning rather than wording, against an authored answer key, marking it faithful, partial, an honest abstention, or invented (a meaning the key does not support). The score reported is the fraction judged faithful. The whole assessment is run with a strong, a middling, and a weak consumer, so the result is read across a capability ladder rather than at a single point. Figure 3 reports the result on the hardest material used in the study: concepts drawn from two unrelated domains, with no shared vocabulary to lean on.

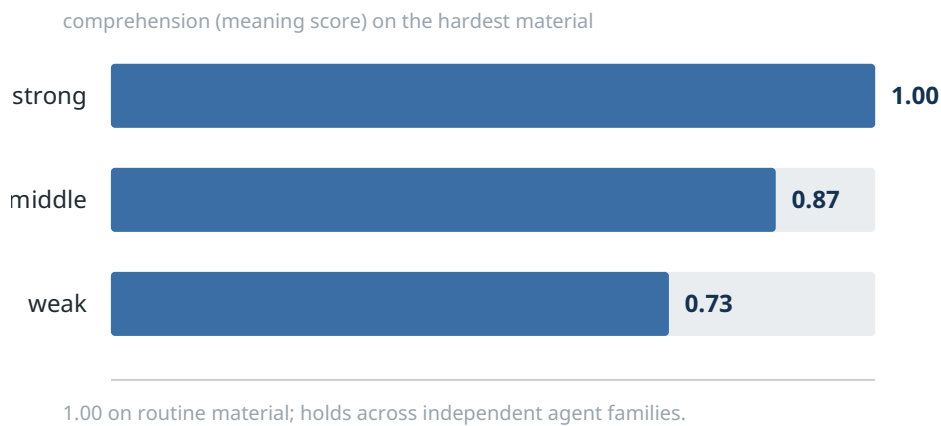


Figure 3: Comprehension of a well-lifted model, from the package alone, by agents of decreasing capability.

Three findings hold, and the numbers are worth stating. First, portability is real. A well-lifted model is understood cold. On routine material comprehension sits at the ceiling for every agent, strong, middling, and weak alike (a comprehension score of 1.00 across the capability ladder). Second, it degrades gracefully rather than abruptly as the reading agent's capability recedes. On the hardest cross-domain material the score is 1.00 for the strong reader, 0.87 for the middling one, and 0.73 for the weak one: a soft gradient, not a cliff. Portability is therefore nearly, though not entirely, consuming agent capability-independent. Third, it is a property of the agentic model capability, not the agentic model lineage. A capable reader from a different agent family reads a lift about as well as one from the family that produced it (a comprehension score of 0.98, against 0.98 for the home strong tier), and it beats the home weak tier. The comprehension judge does not favor its own family: freezing the answers and grading them with two independent judges returns the same score to three figures (0.900 against 0.900).

The one honest floor is that a sufficiently weak agent can confabulate, reporting an understanding it does not have. A weak open model scored 0.56 cold, and its per-trial results were bimodal, either nailing a case or missing it entirely. Portability is thus a claim about capable readers, and, as much, about the attributes of a soundly lifted semantic model. The two go together: the property lives in the model, and a capable reader is what reads it out.

Portability is also a property of the lift, not only of the reader. What the lift works from is the source *surface*: the raw material a system exposes, its labels, its kinds and structural relations, and its instance data, before any meaning is made explicit. A *meaning-poor* surface is one whose labels and structure carry little recoverable sense, for instance opaque identifiers, or names stripped of the definitions that would say what they denote. Such a surface can break portability. Holding the lifting agent fixed and lifting from progressively poorer surfaces, comprehension falls from a solid-lift baseline of 0.94 to between 0.64 and 0.74 when the meaning-bearing surface is stripped away. One finding here is sharp. A surface that keeps a concept's name but strips its meaning is no safer than an anonymous one, because a bare name invites the lifter to confabulate a sense for it. A reference consulted during the lift repairs most of that loss, restoring

the score to between 0.79 and 0.85. This is the first of the reference's two roles. It supplies denotation the source data leaves too thin to lift well. The role is developed, alongside the second, in [Section 7](#).

4.3. Placement of Cognition: the Lift and the Un-lifted Baseline

Two variables underlie the lift and locate it against the alternatives. The first is access: whether the cognition is situated, able to consult the running system, its instances and live state, or un-situated, holding only the published artifact. The second is product: whether the cognition keeps its understanding internal, for its own use, or externalises a self-standing model for others to consume. The lift is the situated and externalising case. [Figure 4](#) places it against the three others.

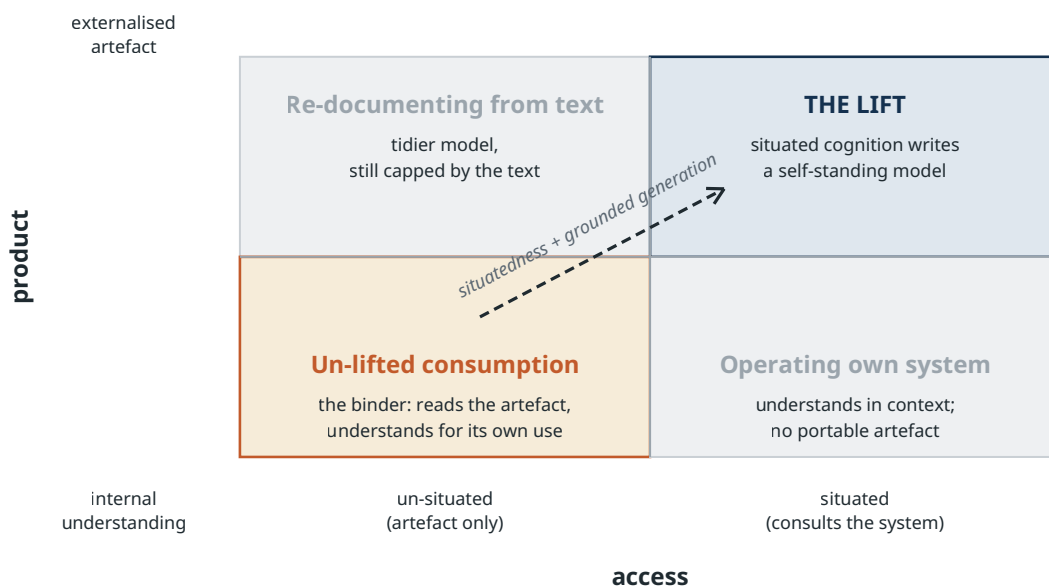


Figure 4: Cognition placed by access and by product.

The contrast that matters is the diagonal. Un-lifted consumption, the un-situated reader that has only the artifact, measures that artifact's portability without the lift; run over a model that was never lifted, it is the un-lifted baseline, and a companion synopsis of related work records exactly this case, an agent binding published descriptions to a thin reference with no lift. The lift is the opposite corner, where a situated cognition writes the meaning down so that such a reader succeeds. What it adds along the diagonal is access, the reach into the system, and grounded generation, the writing-down of what it finds. The remaining corners isolate each half: a situated cognition that does not externalise is an agent operating its own system, understanding in context but leaving nothing portable; an un-situated cognition that does externalise is re-documentation from the text, which tidies but stays capped by what the text holds.

This sharpens the limit of the lift. The place a static artifact defeats an un-situated reader, the fact its text never records, is most often not a true frontier but a gap the lift would close, since the fact is present in the system and a situated lift reaches it. What a concept needs then divides in three: what is implicit but present, made explicit by grounded generation; what is present but elsewhere in the system, supplied by situated access; and what is neither, the residue of authority and underdetermination, referred to a person.

The quadrangle and the cognition spectrum ([Section 6.4](#)) are complementary. The spectrum grades one of the quadrangle's axes, access: where the quadrangle asks whether a cognition is situated or holds only the artifact, the spectrum makes that a dial and splits it per party, from both live through one inert to both inert. It has no axis for the quadrangle's other dimension, product, which is whether the act externalises a self-standing model or keeps its understanding internal, and that is the distinction that separates the lift from mere comprehension. So the two do not collapse into one: the quadrangle names which act is being performed, and the spectrum grades how far that act reaches as cognition recedes. When access is applied to reconciliation, the access that matters is to the other system, which is what the spectrum's placements grade.

5. Pragmatics: Nature, Handling, and Why It Carries the Meaning

Of the parts of a semantic model, pragmatics is the one least talked about. That is probably because it does not freeze cleanly into a schema the way the others do. As it seldom features in schema, it tends to fall out of view and out of consideration. It is, nonetheless, important in operations, and worth setting out plainly. Completing information for a cognitive consumer is, in large measure, capturing the pragmatics a data model leaves out.

Where semantics fixes what a term denotes, pragmatics fixes how the denoted thing is to be used. A semantic model records, as first-class content and over and above the ontology and lexicon, matters such as these:

- whether a figure is a request to be met or a measurement to be reported, an intent versus an observation;
- whose judgment governs a contested field: which side of a boundary owns a route, a grade, or a policy;
- the operative state a value describes: whether it is what was intended or what is operationally the case;
- the conditions under which a signal changes meaning: an alarm during a maintenance window, or a reading when a detector's confidence is low.

Two things about pragmatics emerged clearly from the study. The first is that pragmatics must itself be lifted. It does not survive in the data model, and it is precisely the part a human engineer supplies from outside the artifact. A lift that captures ontology and lexicon but omits pragmatics produces a model that looks complete and is not, and the omission surfaces only when the model

is used to decide something. The second is that pragmatics is often a highly *load-bearing* layer. In a range of operational tasks the meaning of a signal is carried more by its pragmatics than by its structure or its denotation.

Observability is the sharpest case, in the sense of demonstrating both the importance of pragmatics and how it must be handled. The same rising measurement is a matter to act on during normal operation, benign inside a maintenance window, and a matter merely to watch when the detector's confidence is low (Figure 5). None of that is in the value or its type. It is entirely in the pragmatics. The dangerous confusion in this domain is itself a pragmatic one. An *alarm*, a declared undesirable state, is not an *anomaly*, a deviation that may mean nothing. Treating them as the same term is a category error that a purely semantic reading does not catch [RFC9940]. The study bears this out with numbers. When the pragmatic content was carried, the strong and middling agents returned the correct significance verdict (accuracy 1.00 and 0.83) and the false-page storm disappeared. When it was withheld, the same agents paged nearly everything (accuracy 0.17 and 0.08). The verdict was not recoverable from structure alone.

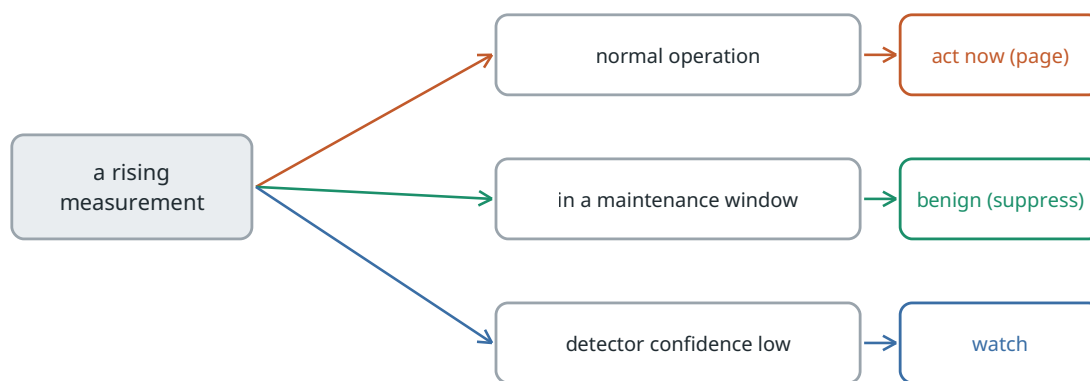


Figure 5: Pragmatics fixes the meaning: one signal, three operating contexts, three verdicts.

Intent shows the same layer deciding a different kind of question. A declarative expectation, such as a lower bound on capacity, an upper bound on latency, or an availability target, is satisfied by a concrete realization rather than equal to it. Which realisations satisfy it, and how conflicts among expectations are traded off, is pragmatic content, not structural. In the study the pragmatic judgment carried the negotiation. A provider returned a best-achievable offer, and the consumer's own declared priorities accepted a degraded value, with no human in the loop (decision accuracy 1.00 for the strong and middling agents).

One further finding sharpens where the payoff sits. Pragmatic content is not uniformly easy to apply. Handed the identical annotations, the weak agent barely moved on the significance verdict (accuracy 0.53 with the content, 0.50 without it). Where the pragmatic content enters as a structural input, however, even a weak agent applies it: given a resource-dependency map, every agent folded a spray of cross-layer symptoms into a single correctly rooted incident, and without that map every agent failed. Judgment is capability-gated; a structural input is not.

Because pragmatics is content, it also diverges between two models and must be reconciled like any other layer. Indeed it is one of the principal axes of divergence in [Section 6](#). The practical consequence is a design instruction. A lift that omits pragmatics is not merely incomplete but silently so, and the layer that most needs deliberate capture is the one a data model is least equipped to hold.

6. Reconciliation: Bridging Two Models by Cognition

A semantic model is an object, and reconciliation is a flagship use of it. Reconciliation aligns two independently built models, so that the systems that use them natively can interoperate. It is a distinct operation with its own objective, bridging the genuine divergence between two adequate models. It is not about patching bad lifts. Two sound models still diverge, and bridging that divergence is the task.

To see the shape of that task, recall what a lifted model contains. It contains a knowledge graph: a schema of typed concepts and permitted relations (the TBox), and the individuals that populate them (the ABox). Over that graph sit the pragmatics and the provenance. Reconciliation must bridge the divergence between two such models wherever it arises: in the ontology, among the instances, and in the pragmatics. These differ in kind, and a treatment that runs them together will miss half the work.

6.1. Ontological correspondence, instance matching, and pragmatics

Ontological correspondence establishes how the two schemas correspond, concept to concept and relation to relation. Two models diverge here on **lexicon** (the same concept under different names, or different concepts under one name) and on **structure and span** (what each treats as an entity, and how finely). One side may bundle into a single concept what the other separates into several. A single legacy alarm, for example, corresponds one-to-many to a separated alarm-state and the fault it implies. Ontological correspondence is a type-level judgment, and where the relation is a refinement rather than an identity it must be confirmed against the specific relation that holds: a bound is *satisfied*, not equalled; an underlay is *ridden*, not renamed.

A third kind of divergence is span itself: a concept one model carries and the other simply lacks, with no counterpart at any cardinality. This one-to-zero case differs from a bundled or refined correspondence, where the content is present on both sides and only carved differently; here it is present on one side and absent on the other, because the two models cut the world at different joints. One side reasons in services while the other holds only the resources a service is realized over. Whether the gap matters depends on the exchange. Where the absent concept is not needed, the honest outcome is to leave it uncovered: a capable agent returns it unmatched rather than inventing a partner. Where it is needed, two disciplined routes supply it. The first is to derive it, since the gap is usually a seam between related things: the deficient side can often synthesize the missing concept from what it does hold, as a service is recovered as a binding over the resources that realize it. The second, where derivation is unavailable, is to extend the shared ground rather than either native model, adding one entry to the ad hoc reference both sides bind through; that keeps the addition disposable, whereas writing the concept into a system's own

schema is the heavier re-documentation this document sets aside. The constructed-reference case of [Section 6.3](#), an IP service meeting a transport circuit, is exactly such a seam and shows the second route in use.

Instance matching, also called entity resolution or co-reference, aligns the individuals once the concepts correspond: which live circuit on one side is which on the other. This is a knowledge-graph-level judgment over keys, attributes, and topology. Most pairs are settled from the models alone. Some are genuine look-alikes, structurally identical on paper, that no static description can separate. Those are settled only by acting on the live system, a point [Section 6.2](#) takes up.

Pragmatics diverges too, and it reduces to neither of these. It is not the ABox. An instance's current state is *observed*; whether that state warrants action here and now is *judged*. So pragmatic divergence is settled not by matching but by resolution: by live elicitation of a party's judgment, or by a pre-placed movable policy that carries that judgment to where it is needed. As cognition recedes, pragmatics is the hardest layer to recover, precisely because it is resolved against a live context rather than stored in the graph.

6.1.1. Disciplined binding

The first thing capable cognition does, with no shared reference at all, is bind what clearly corresponds and refuse what merely looks alike. The discipline is the point. A planted false cognate is refused, not bound. Consider a transport *grade* that is a protection class, set against an IP *grade* that is a class of service: the same word, unrelated meanings. A capable agent declines the pairing on non-lexical evidence, the concepts' kinds, attachments, and instances, which a matcher keying on the shared label cannot do. A capable agent left without a reference commits little, but what it commits is correct. It defers rather than guesses. This is the honest behavior, and it is what makes the resolved fraction and precision separate measures, both worth reporting.

6.2. Verification: confirming a correspondence

Reconciliation does not stop at proposing correspondences. It confirms them, as a step of its own, and keeps only what it can confirm. There are three ways to confirm, and the study exercised all three. How much of this verification is available depends on where the cognition sits. The three ways below describe it at full reach, with both reconciling systems live and able to reason. As cognition recedes from one side or both, less of it can be brought to bear, which [Section 6.4](#) develops as the cognition spectrum.

The first is **question and answer between live agents**. Each side is the authority on its own model, so whatever is ambiguous the other can ask about and get an authoritative answer. In one recorded run, two agents reconciling a cross-domain seam asked and answered one question per concept before either proposed a single binding.

The second is the **decisive virtual experiment**. Because reconciliation operates on the models rather than the live network, an agent can provision a candidate correspondence in virtual space, operate it, and read back whether the invariants a correct translation must preserve still

hold. The question is then settled by operation, not by argument. In a recorded configuration run, one candidate was provisioned and confirmed as an identity, while another was provisioned and refuted, both sides then marking those concepts as having no counterpart.

The third is **round-trip and invariant checking** for an identity correspondence, and a **satisfaction check** for a refinement, where a value cannot be round-tripped because the relation is lossy. Verification is what drives precision toward 1.00 across the spectrum: it rejects the false cognates that slipped into a proposal, and it defers the correct-but-unconfirmable pairings rather than asserting them.

6.3. Building the shared ground when none exists

Where the two models share no reference and none is supplied, the agents can construct a thin shared reference themselves and bind through it. This is the hardest case: two private systems with no public standard between them. Concretely, one such case set a home-grown transport system against a home-grown IP/VPN system, whose worlds touch at a single seam where an IP service rides a transport circuit as its underlay. Making one order across that seam requires binding the circuit to the underlay, binding the transport hand-off to the IP attachment, and pinning three requirements so they cannot be misread: a committed payload rate that is not the bearer line rate, a latency bound that is not a measured value, and protection against a path failure rather than an IP scheduling class. It also requires refusing one look-alike, the transport and IP *grade*.

A reference for this case is deliberately small. It carries one entry per shared seam category, each fixed by a canonical example both sides can instantiate. It does not cover either side's native concepts, and that omission is deliberate: with no *grade* entry to bind to, the cross-domain grade false cognate is pre-empted at the reference itself. The listing below shows the container and two of its five entries.

```

{
  "id": "ref.crossdomain.adhoc.v1",
  "kind": "lexical",
  "reference_type": "ad_hoc_by_example",
  "note": "Constructed between these two agents for this exchange,
    not adopted from any standard. One entry per shared seam
    category, each fixed by a canonical example. It does NOT cover
    either side's native concepts (grade, bearer, vlan, service);
    that omission pre-empts the cross-domain 'grade' false cognate.",
  "entries": [
    { "id": "transport-underlay",
      "label": "transport underlay",
      "synonyms": ["underlay circuit", "carrier circuit"],
      "class": "seam",
      "definition": "The transport circuit that carries an IP
        service as its underlay: the one object both domains
        share, owned by transport and consumed by IP.",
      "example": "circuit CIR-7 is the underlay of service SVC-42" },
    { "id": "service-rate",
      "label": "committed service rate",
      "synonyms": ["committed rate", "CIR"],
      "class": "requirement",
      "definition": "The committed client payload rate the service
        guarantees, NOT the bearer line rate.",
      "example": "5 Gbit/s committed (not the bearer line rate)" }
    // + 3 more: demarcation, latency-bound, protection-requirement
  ]
}

```

Figure 6: An ad hoc reference constructed for one cross-domain seam (container and two of five entries).

Each entry is thin: a stable identifier, a preferred label and synonyms, a shallow class, a disambiguating definition, and one canonical example. It carries no relationship axioms, no cardinalities, and no attribute schemas. It is a lexicon, not a model of the domain. [Figure 7](#) reports what such a self-built reference buys for the strong agent on this case. Left alone with no shared reference, the resolved fraction is 0.20: the agent refuses to guess the seam and so commits little, at full precision. Constructing the thin reference and binding through it lifts the resolved fraction to 0.80. A single decisive virtual experiment on the remaining candidates closes most of the rest, reaching 0.90, against a reference-given ceiling of 1.00. Precision stays at 1.00 throughout, and no false cognate is taken. The climb is in what the agents surface, not a trade against correctness.



Figure 7: Two private models, no shared reference: the agents construct one and bind through it (strong agent; resolved fraction, precision 1.00 throughout).

A resolved fraction below 1.00 here is deferral, not failure. It is what a thin, self-built reference has not yet surfaced, and what the reconciliation's own verification then closes. The reading that matters is that the agents need no standard handed to them. Where one is absent, a capable agent can build the ground the reconciliation stands on. The full role of the reference, and its economics, are taken up in [Section 7](#).

6.4. How far reconciliation reaches: the cognition spectrum

How far a reconciliation can proceed on its own depends on where the machine cognition sits relative to the two systems. This is the master variable, and it governs what a reconciliation achieves, what it costs, and what it can verify. It runs along a spectrum ([Figure 8](#)): both sides cognitive, one side inert, or cognition applied only in overlay to two inert systems.

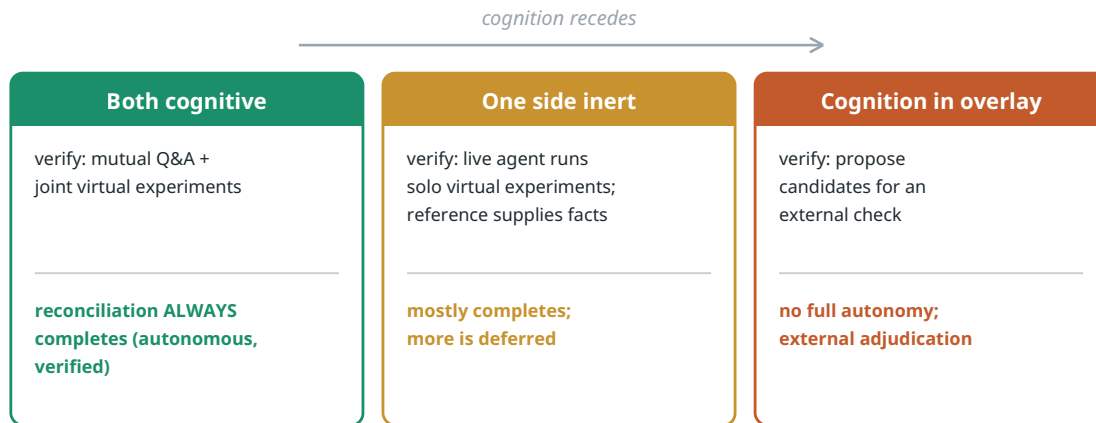


Figure 8: The cognition spectrum: where cognition sits governs what a reconciliation can verify and how far it completes.

At the fully cognitive end, both sides can reason, and here a strong claim holds for sufficiently capable agents. **Reconciliation always completes.** This is an in-principle fact, and in the study it was also verified. It holds for two reasons, both of them functions of live cognition on each side. First, the agents can exchange unbounded further information: each is the live authority on its own model, so whatever is ambiguous the other can ask about and get an authoritative answer. Second, the agents can run decisive virtual experiments, since reconciliation operates on the models rather than the live network. Any question is therefore confirmed, refuted, or authoritatively decided. Across the four scenarios the study found no semantic gap that two sufficiently capable agents could not close, including on the two operations that look least automatable: a two-sided intent negotiation, and the significance verdict in observability. What remains after everything has been exchanged and every experiment run is never an unbridgeable correspondence. It is a concept with no counterpart (correctly returned as unmatched), a fact not yet realized in the running network (an absence in the world, not in meaning), or one of the two irreducible residues discussed in [Section 9](#). None of these is a gap cognition gets stuck on.

As cognition recedes from one side, reconciliation degrades honestly rather than breaking. With one side inert, the live agent can probe the static description but cannot interrogate a peer or co-design an experiment. It reconstructs the inert side's meaning from structure and instances, and a thin reference buys back the facts the inert side cannot supply. Pragmatics is the hardest layer to recover here, because it was resolved against a live context that is now gone. The effect is measurable and its direction depends on capability. For a strong agent a reference is an effort substitute: given the anchor, its hidden deliberation on one schema-binding task collapsed by a factor of about 3.7 with both sides live, about 6 with one side inert, and about 2.4 with both inert. For a weak agent facing an inert side the same reference can add effort rather than save it.

At the far end, cognition sits only in overlay to two inert systems. It can reconstruct both from structure and data, but it can only *propose* correspondences for external adjudication. Full autonomy is not reached. That boundary is real, and it is stated as such in [Section 9](#). The through-line across the spectrum is one claim, made concrete on measured curves: it is cognition that completes a reconciliation, and where cognition is present in full, the reconciliation completes.

7. The Reference, and Its Two Distinct Roles

A shared reference recurs throughout the account, and it earns its place in two separate ways that are easily conflated. Stating them apart is the point of this section.

In its first role, a reference **supports the lift**. Consulted while a model is being produced, it supplies denotation the source surface is too thin to carry, so that a meaning-poor data model can still be lifted into a comprehensible one. This is the repair measured in [Section 4](#), where a reference used in the lift pulled comprehension on a stripped surface back from the 0.64 to 0.74 range up to the 0.79 to 0.85 range. This role serves comprehension generally.

In its second role, a reference **supports reconciliation**. It is the shared common ground two models bind through. Because each side binds its terms to the reference, genuine correspondences arrive as near-identity rather than as candidates to be argued, and false cognates are pre-empted, since two terms that merely share a label bind to different entries. The residual work is confined to what the reference does not cover. This role serves an operation between two parties.

These are genuinely separate roles. A careful lift can make a model comprehensible with no reference at all, so the first role may fall away entirely, while the operations that later use the model can still benefit from the second. The practical conclusion is therefore to make the reference, or a pointer to it, part of the model package regardless. It is thin, and may prove useful to downstream operations.

Its needed nature follows from both roles and is deliberately minimal, as [Figure 6](#) showed. We find a sound reference to comprise a lexicon with thin supports. For each entry it carries a stable identifier, a preferred name and synonyms, a shallow class, a disambiguating definition, and a single canonical example. That example is explanatory, an aid to a reader, not an instance; the reference holds no ABox of its own. It carries no relationship axioms, cardinalities, or attribute schemas. It is not a model of the domain, and it is emphatically not a universal ontology all parties must accept. In practice it is most often derived from an existing model or standard, and, as [Section 6.3](#) showed, where none exists capable agents can build one. One caution the study makes precise: what binds correctly is the reference's *description*, never a bare shared pointer. A shared identifier with no description behind it is worse than no reference at all, because an agent binds by an opaque token and binds wrongly.

The usefulness of a reference is also **graded**. It is load-bearing only where there is no standard to lean on and the agent is capable enough to exploit it. Where a public standard already supplies the common ground, a further reference adds little. Where the agent is too weak to reason over the reference, it cannot rescue the outcome. This shows cleanly on the observability ontological

cognate: the RFC 9940-anchored reference rescued the middling agent completely, driving the surviving cognate from full survival to none, while the weak agent conflated the terms with or without it, and the strong agent never needed it. Between those limits the reference does real work, and it does so cheaply.

That cheapness has a structural source. Because each system binds to a shared reference rather than to every other system pairwise, the operational step count associated with reconciliation grows with the number of systems rather than with the number of pairs. N systems that each bind once to a shared reference replace on the order of $N(N-1)/2$ pairwise translators, and the study verified this linear-versus-quadratic scaling by construction to twelve systems. Supplying an apt reference also reduced the cognitive effort a reconciliation spent, measured in reasoning tokens, by a large factor on the harder scenarios, because the agents argued far less.

8. Evidence from Four Scenarios

The claims above rest on a study run across four network-management scenarios. The scenarios were chosen so that together they span a range of operation classes and the principal ways two models diverge. Each was built, run against a validated gold standard, and reproduced on independently constructed cases. The method is summarised here only as far as is needed to read the results.

8.1. The four scenarios

Configuration

Aligning two standard-governed models of one network, for instance an ONF Transport API view and an IETF traffic-engineering topology [RFC8345] [RFC8795]. Both sides carry public standards the agents already know, so cognition can lean on recognition. This exercises lexical and structural divergence with a standard present. For the strong agent it yields a perfect, verified reconciliation, with precision and resolved fraction of 1.00.

Intent

Negotiating a declarative, quantitative intent into a concrete realization, and carrying operational state back up as fulfillment. The mapping is a refinement: an expectation is satisfied by a realization, not equal to it. The case turns on pragmatic judgment and two-sided negotiation, which completed autonomously at the cognitive end (decision accuracy 1.00 for the strong and middling agents).

Standard-free

Provisioning across two home-grown, private models of adjacent domains that meet at a single seam, with no public standard beneath either side. A transport system and an IP/VPN system whose worlds touch where a service rides a circuit. This is the hardest case, and the one in which the agents must build the shared ground themselves (Section 6.3).

Observability

Reconciling two ways of describing what is wrong with a network. An incumbent fault model that bundles alarm, state, severity, and cause, against a newer model that separates anomaly, symptom, fault, problem, and incident [RFC9940] [I-D.ietf-nmop-network-anomaly-architecture]. This is the scenario in which pragmatics carries most of the meaning (Section 5).

8.2. Cases, agents, and metrics

Each scenario was run on three independently built cases: three different pairs of semantic models to reconcile, deliberately varied in domain, vocabulary, and planted traps, so that a finding resting on one case could be tested against two more. The agents spanned three independent cognitive-agent families and two capability tiers, so that a result could be checked against the possibility that it was an artifact of one model lineage. Outcomes were scored against a validated gold standard for each case. The two headline measures are the resolved fraction and precision defined in Section 2. Cognitive effort was measured in reasoning tokens.

8.3. Reproduction and invariance

Each scenario's signature result reproduced on its independently built cases and held across the agent families and capability tiers. Table 2 summarises this, scenario by scenario.

Scenario	Signature result	Reproduces across
Configuration	Capable agents reconcile two standard models on their own (precision and resolved fraction 1.00 for the strong agent); a thin reference mainly prevents the weaker agents' errors.	3 cases; families; tiers
Intent	Refinement and two-sided negotiation complete autonomously at the cognitive end (decision accuracy 1.00); accuracy over the full multi-step lifecycle falls with agent capability (1.00, 0.88, 0.62).	3 cases; families; tiers
Standard-free	Without a shared reference the strong agent under-commits at full precision (0.20); a constructed reference completes the close (0.80, then 0.90 with a decisive experiment).	3 cases; families; tiers
Observability	The alarm-versus-anomaly false cognate is refused, and the significance verdict is correct once pragmatics is carried: verdict accuracy for the strong and middling agents is 1.00 and 0.83 with it, 0.17 and 0.08 without.	3 cases; families; tiers

Table 2: The signature of each scenario, reproduced across independently built cases and agent families.

Two invariances are worth stating on their own. Every headline result reproduced across the independently built cases, different model pairs returning the same finding, so the results are not a property of one hand-built example. And the headline results held whichever agent family did the reconciling, so they are not an artifact of one vendor's reasoner.

One caution belongs beside the headline numbers, because it bears on acting on a result unsupervised. The dangerous failure is a confident wrong answer, and it concentrates at the weak end and the inert placements. The rate of confident errors rose about elevenfold down the capability ladder, from 0.004 for the strong agent to 0.046 for the weak one, and calibration worsened in step. An agent's own confidence is a weak guard: confidence on wrong bindings sat in the 0.73 to 0.85 range, only a little below confidence on correct ones. A downstream system cannot separate right from wrong by the confidence number alone. The discipline a weak or inert-facing reconciliation needs is an explicit abstain path, a scored declaration of insufficient evidence reported apart from honest deferral, rather than a confidence cutoff.

9. What Holds, and Its Limits

Three conclusions about the concept are supported by the evidence.

Semantic-model portability is real, and it is measurably verifiable. A capable cognitive agent understands and uses a soundly produced semantic model from the package alone. This can be checked directly, across a capability ladder and across independent agent families, rather than inferred from a downstream task. Portability is a property of the model, and, as much, of the reader's cognitive capability.

Capable cognition reconciles divergent semantic models on its own, with discipline. Where the systems reason well enough, two divergent models are reconciled through the agents' own questioning and virtual verification, with no standard supplied. True correspondences are committed and false cognates refused. At the fully cognitive end this is not merely frequent but complete: reconciliation always completes, an in-principle assertion that the study verified.

Portability and reconciliation are graded, not all-or-nothing. Both track the cognition available, degrading gracefully as it recedes rather than collapsing. A thin reference can then partly compensate, rescuing a cognitively weak consumer and making a reconciliation cheaper, though only within the bounds [Section 7](#) describes.

The limits are equally real and are stated plainly. Two residues do not yield to cognition, because they are not questions of fact. One is **authority**. When two sides genuinely conflict over whose value governs a contested field, a reference can supply information but never authority, and the decision stays with a human or an owning party. The study drew this line sharply in the intent scenario: publishing an invariant floor lifted a blind agent's satisfaction check from 0.29 to 0.71, but at both sides inert no reference could move a negotiation whose missing ingredient was the customer's own judgment. The other residue is genuine **underdetermination**. Where no fact settles a correspondence, an honest agent defers rather than invents one. A reconciliation that reaches these residues and defers is behaving correctly, not failing.

And there is the **autonomy boundary** of [Section 6.4](#). Full, unsupervised reconciliation is available at the fully cognitive end. It is not available where machine cognition is employed only in overlay to two inert systems, which can only propose correspondences for external adjudication. Between those ends the automation degrades predictably, and a thin reference buys back part of what receding cognition gives up.

10. Implications for Network Management and Standardization

If much of what was formerly pre-agreed can be produced and reconciled on demand by cognition, the question for the community is not whether to abandon standard models but whether such models should be evolved, and whether it might be helpful to introduce other standardized artifact types. The evidence points to an evolution rather than a rupture.

What may be most useful to agree in advance may narrow and simplify. As models can increasingly be produced and reconciled ad hoc, for the task at hand, the heaviest fully-specified schemas may give way to lighter, more broadly usable artifacts. Thin shared references, carrying names, definitions, and examples derived from existing models, are the anchor to which independently built systems bind. Such references are cheaper to agree and cheaper to maintain. They are also precisely the kind of interoperable anchor to which emerging knowledge-graph and AI-modeling efforts in the community could bind. The community's knowledge-graph work for network operations [[I-D.mackey-nmop-kg-for-netops](#)] meets this approach squarely, since a lifted semantic model contains a knowledge graph a machine can consume, and a thin shared reference is exactly the kind of anchor such graphs can bind to.

What still needs agreeing is narrower than a full shared model, and it is of two kinds. The first is the thin references themselves: the small shared anchors reconciliation binds through, worth agreeing precisely because they are cheap and broadly reusable. The second is the authority boundaries the study keeps reaching ([Section 9](#)): the contested fields where two systems hold conflicting values and no reasoning can decide whose governs, because the question is one of authority, not of fact. This second kind is not a modeling problem, and better cognition will not dissolve it. Agreeing who governs what is a matter of governance, and it is exactly where standardization remains indispensable, even as ad hoc reconciliation takes over much of the rest.

A further implication follows from where the model now comes from. A semantic model is produced by a system's own cognition, the lift, and its form is flexible: what matters is not a fixed schema but that the model carries the elements this document describes, its ontology and lexicon, its pragmatics and its provenance, and explains itself well enough for a capable agent to pick it up. The burden that once fell on a shared model therefore shifts onto a system's ability to describe and interrogate itself. Where a system exposes enough of its structure, its local concepts, its current facts, and its authority and provenance, a consumer's cognition can lift and reconcile against it ad hoc. Where a system exposes none of that, and the meaning is neither in its schema nor available elsewhere, autonomous reconciliation cannot be guaranteed, and the missing content must be drawn from another authoritative artifact, elicited from a live interface, or supplied by a person.

This points to a second candidate for standardization, alongside the thin references: not a universal domain model, but a system's self-description and interrogation. What is worth agreeing is how a system exports its structural surface, explains its local concepts, makes its current facts discoverable, declares its authority and provenance, and answers open questions from a consumer that does not know its schema in advance. One architectural pattern that realises this is an interface that declares the concepts and relationships a system supports and assembles the requested knowledge at runtime, rather than fixing all exchanged content at design time, carrying its own provenance and confidence and able to defer where the evidence is insufficient. The point is the capability, not any one serialization or interface: agree how a system explains itself, not the model it uses inside.

None of this displaces the installed base. Ad hoc production and reconciliation coexist with legacy standards and legacy systems, machine to machine, for the task at hand, while the standardization effort is redirected toward the anchors and the authority boundaries that actually need agreeing. The direction is an evolution of where the community spends its modeling effort, not a repudiation of the value that pre-agreement has delivered.

11. Related Work

Two earlier Internet-Drafts introduced, at an earlier and less mature stage, ideas this document develops and tests. The first, by one of the present authors, took up agent-to-agent reconciliation of the ontologies implicit in data models [[I-D.janz-nmrg-ontology-reconciliation](#)]. The second, by both present authors, introduced thin shared reference lexicons as the anchor for such reconciliation [[I-D.janz-nmrg-reference-lexicons](#)]. A third draft, again by one of the present authors, treats the consumer-provider relationship in Network-as-a-Service as a continuous agentic negotiation [[I-D.janz-nmrg-naas-agentic-negotiation](#)], the operational setting of the intent scenario here. The present document supersedes the conceptual treatment in those drafts where they differ, and supplies the empirical grounding they lacked.

The work sits alongside efforts across the network-management community that share its premise, that operational meaning must be held as structured, machine-consumable knowledge and not as form alone. The community's knowledge-graph framework for network operations [[I-D.mackey-nmop-knowledge-graph-for-netops](#)] proposes exactly the structured, machine-consumable representation this document lifts data into, and the NMOP work on network fault and anomaly terminology and architecture [[RFC9940](#)] [[I-D.ietf-nmop-network-anomaly-architecture](#)] supplies the vocabulary the observability scenario reconciles against. The network telemetry framework [[RFC9232](#)] and the standard topology and traffic-engineering models [[RFC8345](#)] [[RFC8795](#)] supply further common ground the study draws on. The semantic model, the lift, and the thin shared reference developed here are complementary to all of these. A reference is exactly the interoperable anchor to which such knowledge graphs and modeling interfaces could bind. The theoretical grounding is the classical study of signs [[Morris](#)], with the data-to-wisdom lineage [[Ackoff](#)] as an alternative framing that arrives at the same structure.

12. Open Questions and Next Steps

One important next step is to move the study from purpose-built cases to larger, real, operator-sourced datasets, spanning the same scenarios at production scale and with the messiness of real data. Real models diverge in ways a designed case cannot fully anticipate. The measures reported here, portability, resolved fraction and precision, the reference's graded value, and the cost in reasoning effort, should be re-established on operator data before they are relied upon.

A further step is more consequential, and only an operator can take it: to attach cognition to a live operating system and have it perform the lift in place, with real access to state, instances and pragmatics, rather than over a frozen export. This matters more than more inert data. An inert dataset, however large, fixes cognition at the un-situated corner of [Figure 4](#): it refines the baseline and re-measures the same surface bound, and it cannot separate a fact the lift could recover from the system from one that is genuinely absent, because past the surface everything looks like one wall. A situated lift makes that separation, and what remains unresolved after a maximally situated lift is the true residue, the authority and underdetermination remainder that bounds automation and that this work defines but does not yet measure.

What the experiment involves can be stated concretely. The target is a live management plane with a machine-readable operational view: an ONF TAPI controller, an IETF-YANG datastore reached over NETCONF or RESTCONF under NMDA, a gNMI-served state stream, or a controller such as ONOS or OpenDaylight. The cognition is a capable agent given read-only tools over that plane, datastore reads on the running and operational datastores, topology and inventory reads, a context export where the interface offers one, a telemetry subscription, and a bounded budget of read-only queries it may issue to settle a question, together with its prior over modeling conventions and, optionally, a thin reference. Safety is a first-class constraint: read-only credentials, rate limits, an audit of every call, and no path to a write. Situatedness is a dial the experiment should sweep, from the frozen export alone (the un-lifted baseline), through structural and relational reads, to instances read from the live datastore, to live state, context export and bounded probes, and finally to a self-lift in which the system's own agent authors its meaning.

For a chosen slice of the model the agent proceeds concept by concept: it enumerates the concepts from the schema surface; reads each concept's structure, relations and real instances; where the source gloss is thin or absent, generates a grounded gloss from structure, instances and prior and checks it against them; reads the pragmatics, what the thing is for and whose authority governs it, from a context export, from governance or configuration, or, where only a human holds it, by asking; records provenance, marking each fact's method and firmness; and proposes a reference binding. A verifier pass strikes any generation the material does not support, and every fact the agent can neither ground nor read is flagged as a residue candidate for an authority to settle. Portability is then judged exactly as in [Section 4.2](#), but on a model lifted in situ: the lifted model, and only the lifted model, is handed to an independent consumer, and the headline number is the value of the lift, the portability of the lifted model minus the portability of the un-lifted baseline on the same concepts. Crossing the situatedness dial with this

measure is the core result: as access climbs, portability should rise and the unresolved set should shrink toward the residue, so that what remains at the top of the dial is the first measurement of the irreducible remainder.

Two scenario cases carry the experiment. The first is a configuration seam: a standard-governed model, an ONF TAPI or IETF-YANG view, lifted in situ from a live controller. It is the readiest clean case, where public standards are present, the surface is rich, and portability is the headline a situated lift can establish with least friction. The second is the observability case, and it is the sharper of the two, because the pragmatic layer is decisive there. In the study, the significance verdict, whether a rising measurement is an alarm, a benign maintenance effect, or a low-confidence reading, is recovered only when the pragmatics are present (verdict accuracy 1.00 and 0.83) and collapses when they are stripped (0.17 and 0.08), and the context that fixes significance lives in the running system rather than in any static model. A focused run would attach cognition to an operator's alarm-and-anomaly system, lift its alarm model in situ with the live context, bind it to the shared anomaly-semantics reference, and measure both portability and the pragmatic verdict against the un-lifted baseline. The NMOP work on network anomaly detection and its semantics ([RFC9940] [I-D.ietf-nmop-network-anomaly-architecture]) is already assembling the operational data, the term ladder, and, through its hackathons, the venue such a run would need, and the knowledge-graph representation the data can be held in ([I-D.mackey-nmop-kg-for-netops]) is the form a lifted model naturally takes. The barrier to entry is low, because those inputs and that venue already exist.

The authors invite operators and NMRG and NMOP participants to contribute real model pairs and datasets, to attempt situated lifts in these settings, and to help define a shared benchmark over them.

Beyond scale, several questions the study opens remain. There is lifting from raw schema text rather than a curated surface, and from a cold start with no instances to read. There is the maintenance of a lifted model and of a shared reference over time. And there is the shape of a published reference ecosystem, and how references derived by different parties from the same standard relate.

13. Security Considerations

This document describes a direction and reports experimental findings. It defines no protocol and introduces no new wire format. Nonetheless, several considerations follow from the direction. A semantic model and any accompanying reference become inputs a cognitive agent acts on, and so are targets for manipulation. A poisoned reference, or a lifted model with falsified pragmatics or provenance, could steer a reconciliation to an incorrect but confident result. The provenance layer of a semantic model, who asserted a fact, by what method, and how firmly, is therefore security-relevant and should be integrity-protected, and references should be authenticated to their derivation source. The autonomy boundary of [Section 9](#) is also a safety boundary. Decisions of authority, and correspondences an honest agent defers, must not be silently resolved by an agent acting beyond its competence. This concern is sharpened by a measured finding: agents assert wrong bindings at nearly the confidence of correct ones, so a downstream gate cannot rely on a confidence score alone. Reconciliation outcomes intended to

drive operational change should carry the confidence and provenance on which they rest, together with an explicit declaration where evidence was insufficient, so that a supervising system or operator can gate action accordingly.

14. IANA Considerations

This document has no IANA actions.

15. Informative References

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